

BMI Difference in Endurance and Strength Athletes: Olympic Triathletes vs Weightlifters

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Introduction

Body Mass Index (BMI) relates weight to height and is a proxy for body fat [2]. Because BMI uses only height and weight, it is impossible to distinguish between body fat and muscle mass, therefore in elite athletes who tend to have low body fat it can be used as proxy for muscle mass [3].

Endurance athletes aim to move their own body weight for long distances as efficiently as possible, therefore they tend to have low muscle mass [3]. In contrast, strength athletes aim to apply as much force as possible in a very short moment, consequently they tend to have high muscle mass [3].

It is expected that these differences in body composition will be reflected by different BMI as athletes with the same height will have a different weight due to demands of their sport. For this study, triathlon (combining swimming, cycling and running) was selected as endurance sport, due to its long duration. Weightlifting was selected as strength sport because it mainly focuses on rapid force production. Men and women were studied separately because they differ in body composition.

The research aims to answer the question: *Do Olympic triathletes and weightlifters differ significantly in BMI, how large is this difference for men and women, and does the difference itself differ between the sexes?*

Data sample

The dataset [1] contained information on Olympic athletes from Athens 1896 to Rio 2016. It had 271,116 rows and 15 columns. Each row corresponded to one athlete per event. This study focused on columns containing athlete ID, sex, height (cm), weight (kg), games, year and sport.

The dataset required some cleaning to remove unnecessary columns, athletes with missing entries, invalid BMI or duplicates and to select years with both sports present (2000–2016).

Methods

The dataset was split into four subsamples based on sex and sport. The final sample contained 265 male and 261 female triathletes and 775 male and 459 female weightlifters. BMI was calculated for each athlete as $\mathrm{BMI} = m/h^2$, where m is weight in kg and h is height in m. A mean for each subsample was found, the uncertainty was calculated using formula of standard error of the mean: $\sigma_{\bar{x}} = s/\sqrt{n}$.

For each sex, the difference in mean BMI between the two sports was calculated with uncertainty $\sigma_\Delta = \sqrt{\sigma^2_{\bar{x},\mathrm{WL}} + \sigma^2_{\bar{x},\mathrm{TRI}}}$, and tested with a two-tailed Welch's t-test ($\alpha = 0.01$). To test whether this difference differs between men and women, a z-test was performed. Assuming the samples are independent, the uncertainties of the two differences were combined, giving $z = |\Delta_m - \Delta_f| \,/\, \sqrt{\sigma^2_{\Delta_m} + \sigma^2_{\Delta_f}}$. The difference between the sexes was considered significant if $z > 2.576$ ($\alpha = 0.01$, two-tailed).

Results and discussion

First we look at the density distribution of triathlete and weightlifter BMI per sex. Figure 1 shows that for both sexes triathlete mean BMI is lower than weightlifter BMI. Triathlete BMI is very densely distributed around mean with most values being around 20-23 for men and 18-21 for women. The weightlifter BMI is more widely distributed reaching the maximum of 53.79 for men and 51.79 for women. That can be explained by weight class division - there is no upper bound for heavyweights.

As to comparison per sex, in both sports the mean BMI for men is higher than for women (3.1 kg/m² (WL), 1.6 kg/m² (TR)). The plot representing female BMI is slightly shifted to left illustrating the potential difference in men and women body composition.

The statistical analysis revealed that weightlifters had a significantly higher BMI than triathletes in both sexes. The difference was 7.93 ± 0.22 kg/m² for men ($p = 5.7 \times 10^{-176}$) and 6.38 ± 0.28 kg/m² for women

($p = 8.6 \times 10^{-82}$), both well below $\alpha = 0.01$.

The difference between the sports was 1.5 ± 0.4 kg/m² larger for men than for women. This corresponds to $z = 4.4$, above the critical value of 2.58, so the difference between the sexes is significant.

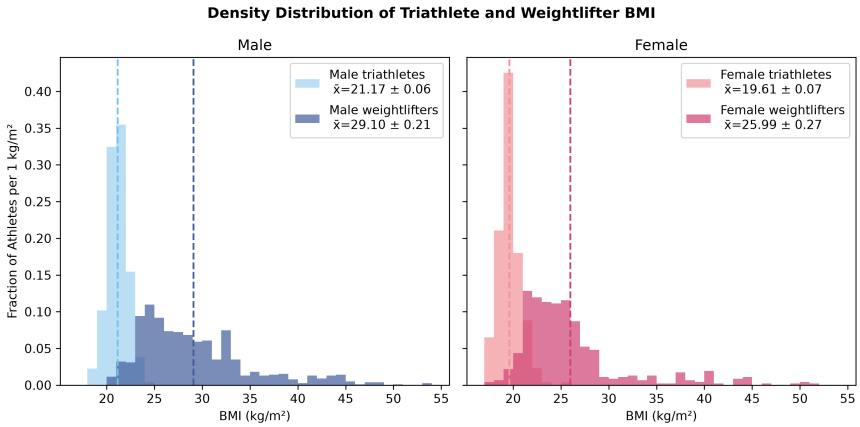


Figure 1: Density distribution of BMI for Olympic triathletes and weightlifters (2000–2016), for men (left) and women (right). Dashed lines show the mean BMI of each group; bin width 1 kg/m².

In conclusion, weightlifters had higher BMI than triathletes in both sexes, this difference was significantly larger for men than for women. This is consistent with predicted lower body mass in endurance athletes and higher body mass in strength athletes [3]. The findings suggest that physical demands of sports discipline and sex might affect athletes body composition.

The study had several limitations. For instance, some athletes were excluded due the missing data on height and weight, weight classes in weightlifting introduced certain clustering. In addition athletes competing at multiple Games were counted more than once making the data non-independent and affecting the uncertainties. Future studies should assess independent athletes and use more direct body composition measurement instead of BMI.

References

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